

Data processing

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First import the table with weights and thickness

```
library(tidyverse)
library(readxl)
library(stringr)
library(janitor)
library(units)
library(emmeans)
library(lmerTest)
library(lme4)
library(performance)
library(see)
library(DHARMA)
library(pbkrtest)
```

Now we start with the data

```
fines_data <- read_xlsx("raw_data/Fines_study.xlsx",
                        sheet = "Handsheet_properties")
```

Extract the pulping data, and just select the kappa number

Here i can select which columns are going to appear in my final table

```
Pulping_data <- read_xlsx("raw_data/Fines_study.xlsx",
                          sheet = "Pulping") #|>
# select(Biomass, Code, Kappa, Total_screened_yield, )
```

Here i can select which columns are going to appear in my final table

```
freeness_data <- read_xlsx("raw_data/Fines_study.xlsx",
                           sheet = "Freeness") |>
  select(ID1, freeness)
```

Pulping_data

```
# A tibble: 6 x 15
  Biomass Code Pulping Chemical_charge Liquor_Biomass_ratio pH Temperature
  <chr> <chr> <chr> <dbl> <dbl> <dbl> <dbl>
1 W W1 Ammonium~ 0.25 6 5.5 150
2 W W2 Ammonium~ 0.25 6 5.5 150
3 W W3 Ammonium~ 0.25 6 5.5 150
4 E E1 Ammonium~ 0.25 4 5.5 160
5 E E2 Ammonium~ 0.25 4 5.5 160
6 E E3 Ammonium~ 0.25 4 5.5 160
# i 8 more variables: Time_at_max_T <dbl>, Raw_material_OD <dbl>,
# Total_pulp_OD <dbl>, Total_yield <dbl>, Screened_mass <dbl>,
# Screening_yield <dbl>, Total_screened_yield <dbl>, Kappa <dbl>
```

Getting the data from the tensile tests

```
tensile_data <- read_xlsx("raw_data/tensile_data.xlsx")
tensile_data
```

```
# A tibble: 105 x 7
  ID `F_[N]` `eF_[%]` `tFmax [s]` `ModElas_[MPa]` `TEA_[J/m²]`
  <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
1 E1N13 37.6 17.6 2.5 39.3 302.
2 E1N12 36.5 17.9 2 38.5 301.
3 E1N1 36.9 19.3 2.5 38.1 331.
4 E2N1 35.8 19.6 2.5 32.0 315.
5 E2N11 35.4 19.3 2.5 34.8 316.
6 E2N12 38.0 21.0 2.5 38.0 377.
7 E3N1 31.7 13.7 2 35.6 191.
8 E3N11 33.4 14.6 2 38.2 219.
9 E3N13 33.8 14.6 2 38.7 220.
10 E1F11 36.3 18.5 2 36.9 310.
# i 95 more rows
# i 1 more variable: `TStif_ISO [kN/m]` <dbl>
```

We need to create the variables for each of the columns

```
all_data <- fines_data |>
  mutate(biomass = as.factor(Biomass),
         Biomass = NULL,
         fines = factor(ifelse(
           str_detect(ID, "F"), "Y", "N")),
         hs_treatment = factor(
           case_when(
             str_detect(ID, "A") ~ "additive",
             str_detect(ID, "5") ~ "recirculation",
             str_detect(ID, "[1-4]") ~ "control"),
           levels = c("control", "recirculation", "additive")),
         Code = str_extract(ID, "^[WE]\\d"),
         ID1 = str_extract(ID, "^[WE]\\d.")
  ) |>
  inner_join(Pulping_data, by = join_by(Code)) |>
  left_join(freeness_data, by = join_by(ID1)) |>
  left_join(tensile_data, by = join_by(ID)) |>
  select(-Biomass, -ID1) |>
  relocate(Code, biomass, fines, hs_treatment, .after = ID) |>
  mutate(
    id2 = str_extract(str_remove(ID, "A"), "^.{3}")
  )
```

Now we get a code to extract the data from the FQA values

```
fqa_data <- read_excel("C:/Users/yacordob/OneDrive - North Carolina State University/Fines Stu
  sheet = "FQAResult",
  range = "A8:B56") |>
  mutate((read_excel("C:/Users/yacordob/OneDrive - North Carolina State University/Fines Stu
    sheet = "FQAResult",
    range = "B8:B56")),
    (read_excel("C:/Users/yacordob/OneDrive - North Carolina State University/Fines Stu
      sheet = "FQAResult",
      range = "B8:B56")),
    (read_excel("C:/Users/yacordob/OneDrive - North Carolina State University/Fines Stu
      sheet = "FQAResult",
      range = "B8:B56")),
    (read_excel("C:/Users/yacordob/OneDrive - North Carolina State University/Fines Stu
      sheet = "FQAResult",
      range = "B8:B56")))
```



```

      (read_excel("C:/Users/yacordob/OneDrive - North Carolina State University/Fines Stu
                sheet = "FQAResult",
range = "B8:B56")),
      (read_excel("C:/Users/yacordob/OneDrive - North Carolina State University/Fines Stu
                sheet = "FQAResult",
range = "B8:B56"))
)|>
  drop_na(SampleId)

```

Now that all the data is in that table, we need to fix the data

```

fqa2_data <- fqa_data|>
  slice(-c(1:13, 19, 22, 23, 27, 29, 31)) |>
  drop_na(E1F1r) |>
  pivot_longer(-SampleId,
               names_to = "Sample",
               values_to = "Value") |>
  pivot_wider(names_from = SampleId,
              values_from = Value) |>
  clean_names() |>
  mutate(across(-sample, as.double)) |>
  mutate(id2 = str_extract(sample, "[WE]\\d[FN]")) |>
  group_by(id2) |>
  summarise(
    across(where(is.numeric), mean, na.rm = T
  ))

```

Warning: There was 1 warning in `summarise()`.
 i In argument: `across(where(is.numeric), mean, na.rm = T)`.
 i In group 1: `id2 = "E1F"`.
 Caused by warning:
 ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
 Supply arguments directly to `.fns` through an anonymous function instead.

```

# Previously
across(a:b, mean, na.rm = TRUE)

# Now
across(a:b, \(x) mean(x, na.rm = TRUE))

```

Setting units

```

area <- set_units(0.02, "m^2")
hyp_data <- all_data |>
  select(c(1:7, 16, 19:23, 25:27)) |>
  mutate(weight = set_units(weight, "g"),
         thickness = set_units(thickness, "um"),
         `F_[N]` = set_units(`F_[N]`, "N"),
         # Total_yield = set_units(Total_yield*100, "%"),
         # Total_screened_yield = set_units(Total_screened_yield*100, "%"),
         `ModElas_[MPa]` = set_units(`ModElas_[MPa]`, "MPa")
  ) |>
  mutate(bw = weight*0.93 / area,
         density = set_units(bw / thickness, "g/cm^3"),
         T_index = set_units(`F_[N]` / set_units(15, "mm") / bw, "kN*m/kg"),
         Code = factor(Code)
  ) |>
  drop_units() |>
  group_by(
    Code,
    biomass,
    fines,
    hs_treatment
  ) #|>
  # summarise(
  #   T_index = mean(T_index, na.rm = TRUE),
  #   n_sheets = n(),
  #   .groups = "drop"
  # )

```

Running for hypothesis test

First we test whether fines affect tensile or not

$$H_o : \mu_{Fines} = \mu_{NoFines}$$

$$H_a : \mu_{Fines} \neq \mu_{NoFines}$$

```

mod_full <- lmer(
  T_index ~ biomass * fines * hs_treatment +
    (1 | Code),
  data = hyp_data
)

anova(mod_full, type = 3)

```

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)	
biomass	58.25	58.246	1	4.014	10.9711	0.0294339	*
fines	76.88	76.878	1	90.022	14.4806	0.0002579	***
hs_treatment	374.25	187.124	2	90.022	35.2464	4.952e-12	***
biomass:fines	53.72	53.722	1	90.022	10.1190	0.0020138	**
biomass:hs_treatment	7.11	3.553	2	90.022	0.6693	0.5145859	
fines:hs_treatment	5.59	2.795	2	90.022	0.5264	0.5925465	
biomass:fines:hs_treatment	18.52	9.259	2	90.022	1.7440	0.1806690	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Based on the p-values, we reject the null hypothesis, and have strong evidence to conclude that the probability of getting this value under the assumption that the null hypothesis is true, is very low. Therefore we conclude that Fines do have an effect on the tensile index of the handsheets.

Also, based on the p-value for the interaction between biomass:fines, we have strong evidence to conclude that there is an interaction between biomass and fines, so the effect differs between eucalyptus and wheat straw.

Based on the p-value for the interaction between fines and the handsheet treatment, we fail to reject the null hypothesis, and conclude that the handsheet treatment does not change the effect of the fines (no direct correlation between the treatment and the fines type)

And finally based on the interaction term between biomass, fines, and treatment, we can conclude that the use of the cationic polymer, or the re circulation does not affect differently the wheat straw of the eucalyptus.

```
hs_data <- hyp_data %>%
  mutate(
    Code = factor(Code),

    biomass = factor(
      biomass,
      levels = c("E", "W"),
      labels = c("Eucalyptus", "Wheat_straw")
    ),

    fines = factor(
      fines,
      levels = c("N", "Y"),
      labels = c("No_fines", "Fines")
    ),
```

```

hs_treatment = factor(
  hs_treatment,
  levels = c(
    "control",
    "recirculation",
    "additive"
  ),
  labels = c(
    "Control",
    "Recirculation",
    "Polymer"
  )
)
)
)

```

```

sample_size_table <- hs_data %>%
  group_by(
    biomass,
    Code,
    fines,
    hs_treatment
  ) %>%
  summarise(
    n_measured = sum(!is.na(T_index)),
    n_missing = sum(is.na(T_index)),
    .groups = "drop"
  )

sample

```

```

function (x, size, replace = FALSE, prob = NULL)
{
  if (length(x) == 1L && is.numeric(x) && is.finite(x) && x >=
    1) {
    if (missing(size))
      size <- x
    sample.int(x, size, replace, prob)
  }
  else {
    if (missing(size))
      size <- length(x)
  }
}

```

```

      x[sample.int(length(x), size, replace, prob)]
    }
  }
<bytecode: 0x00000171f279e698>
<environment: namespace:base>

```

```

cell_data <- hs_data %>%
  group_by(
    Code,
    biomass,
    fines,
    hs_treatment
  ) %>%
  summarise(
    n_handsheets = sum(!is.na(T_index)),

    mean_T_index = if_else(
      n_handsheets > 0,
      mean(T_index, na.rm = TRUE),
      NA_real_
    ),

    sd_T_index = if_else(
      n_handsheets > 1,
      sd(T_index, na.rm = TRUE),
      NA_real_
    ),

    se_T_index = if_else(
      n_handsheets > 1,
      sd_T_index / sqrt(n_handsheets),
      NA_real_
    ),

    .groups = "drop"
  )

```

Descriptive statistics

```

results_summary <- cell_data %>%
  group_by(
    biomass,

```

```

    fines,
    hs_treatment
  ) %>%
  summarise(
    n_cooks = sum(!is.na(mean_T_index)),
    mean = mean(mean_T_index, na.rm = TRUE),
    sd_between_cooks = sd(mean_T_index, na.rm = TRUE),
    se_between_cooks = sd_between_cooks / sqrt(n_cooks),
    .groups = "drop"
  )

results_summary

```

A tibble: 12 x 7

	biomass	fines	hs_treatment	n_cooks	mean	sd_between_cooks	se_between_cooks
	<fct>	<fct>	<fct>	<int>	<dbl>	<dbl>	<dbl>
1	Eucalyptus	No_f~	Control	3	39.1	2.59	1.50
2	Eucalyptus	No_f~	Recirculati~	3	41.0	1.10	0.636
3	Eucalyptus	No_f~	Polymer	3	37.0	1.30	0.752
4	Eucalyptus	Fines	Control	3	37.4	1.51	0.871
5	Eucalyptus	Fines	Recirculati~	3	38.1	0.980	0.566
6	Eucalyptus	Fines	Polymer	3	32.2	1.77	1.02
7	Wheat_str~	No_f~	Control	3	34.4	0.786	0.454
8	Wheat_str~	No_f~	Recirculati~	3	34.4	1.26	0.727
9	Wheat_str~	No_f~	Polymer	3	30.5	3.11	1.80
10	Wheat_str~	Fines	Control	3	33.6	4.09	2.36
11	Wheat_str~	Fines	Recirculati~	3	34.1	2.68	1.55
12	Wheat_str~	Fines	Polymer	3	30.7	2.34	1.35

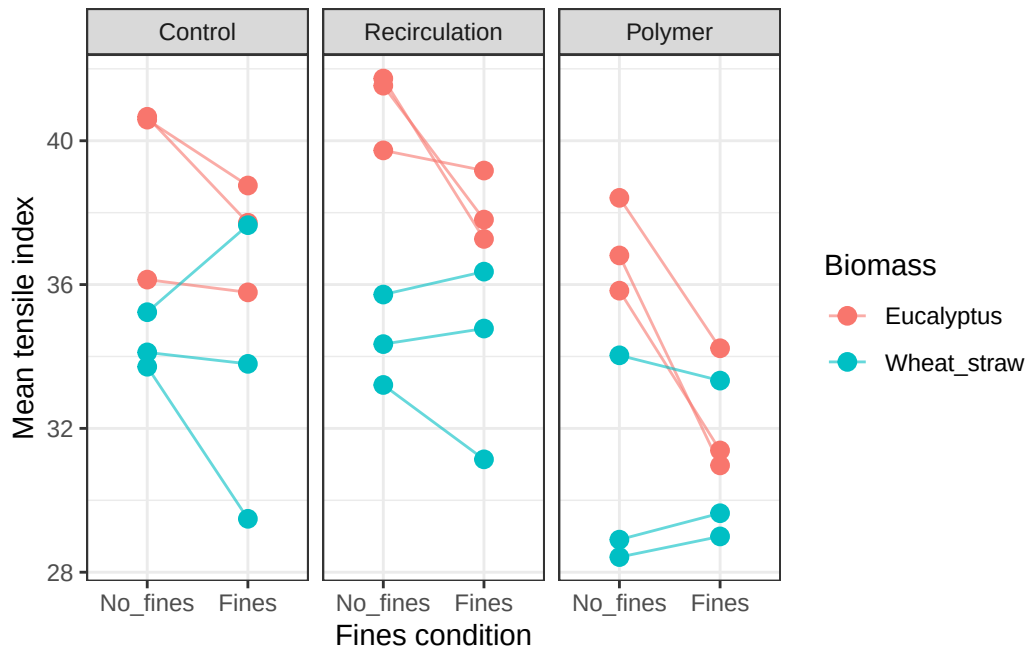
Cook level responses

```

ggplot(
  cell_data,
  aes(
    x = fines,
    y = mean_T_index,
    color = biomass,
    group = interaction(biomass, Code)
  )
) +
  geom_line(alpha = 0.60) +

```

```
geom_point(size = 2.8) +
facet_wrap(~ hs_treatment) +
labs(
  x = "Fines condition",
  y = "Mean tensile index",
  color = "Biomass"
) +
theme_bw()
```



An upward line from No_fines to Fines means fines increased tensile index.

A downward line means fines decreased tensile index.

Different slopes for Eucalyptus and wheat straw suggest a biomass $\times$ fines interaction.

Different slope patterns across the three panels suggest a three-way interaction.

```
mod_cook <- lmer(
  mean_T_index ~ biomass * fines * hs_treatment +
    (1 | Code),
  data = cell_data,
  REML = TRUE,
  na.action = na.exclude
)
```